

Reducing Bias in Predictive Healthcare Models Through Algorithmic Fairness Techniques

Course: DSC-590

Assignment: Model Implementation and Testing

This notebook implements and evaluates a predictive healthcare model while explicitly addressing potential sources of algorithmic bias. Using a diabetes prediction dataset, the project examines model performance, error behavior, and fairness-related considerations to ensure responsible and ethical use of machine learning in healthcare decision-making.

Import required libraries

2. Load and Inspect the Dataset

The dataset used in this project contains anonymized patient health indicators commonly associated with diabetes diagnosis. Because healthcare models can unintentionally amplify bias present in historical data, an initial inspection is performed to understand feature distributions and ensure the data is appropriate for fairness-aware modeling.

```
In [4]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import (
    accuracy_score, precision_score, recall_score,
    classification_report, confusion_matrix, ConfusionMatrixDisplay
)
```

```
In [10]: # Option 1: Specify the correct path to the file
# Replace with the actual path where diabetes.csv is located
data = pd.read_csv("C:\\Users\\jeffm\\Downloads\\New folder (3)\\New folder\\diabet

# Option 2: If you're using a sample dataset from a library like seaborn
# import seaborn as sns
# data = sns.load_dataset('diabetes') # If available in seaborn
# data.head()

# Option 3: Download the dataset from a URL
# import pandas as pd
# url = "https://raw.githubusercontent.com/datasciencedojo/datasets/master/diabetes"
```

```
# data = pd.read_csv(url)
# data.head()
```

Markdown Cell: Features and Target 3. Define Features and Target Variable

```
In [11]: X = data.drop('Outcome', axis=1)
y = data['Outcome']
```

=====

Markdown Cell: Train-Test Split

=====

4. Train-Test Split

Split 80/20 with stratification to preserve class balance.

```
In [12]: X_train, X_test, y_train, y_test = train_test_split(
X, y, test_size=0.2, random_state=42, stratify=y
)
```

Markdown Cell: Feature Scaling

5. Feature Scaling

Standardization improves numerical stability.

```
In [13]: scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

```
In [ ]: ## 6. Model Implementation and Training
```

A logistic regression model was selected for this project due to its transparency and which are critical when evaluating fairness in healthcare predictions. Interpretability

practitioners to better understand how input features contribute to outcomes **and** to potential sources of bias within the decision-making process.

```
In [14]: model = LogisticRegression(max_iter=1000, random_state=42)
model.fit(X_train_scaled, y_train)
```

```
Out[14]: LogisticRegression
LogisticRegression(max_iter=1000, random_state=42)
```

Markdown Cell: Predictions

7. Model Predictions

```
In [15]: y_pred = model.predict(X_test_scaled)
y_pred[:10] # preview first 10 predictions
```

```
Out[15]: array([1, 0, 0, 0, 0, 0, 0, 1, 0, 1])
```

8. Evaluation Metrics

Model performance is evaluated using accuracy, precision, and recall. While these metrics provide insight into predictive capability, they are also important for fairness analysis. Disparities in false positives or false negatives can disproportionately affect certain patient populations, making metric interpretation essential in healthcare applications.

```
In [16]: accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)

print(f"Accuracy: {accuracy:.3f}")
print(f"Precision: {precision:.3f}")
print(f"Recall: {recall:.3f}")
```

Accuracy: 0.714
Precision: 0.609
Recall: 0.519

Markdown Cell: Classification Report

9. Classification Report

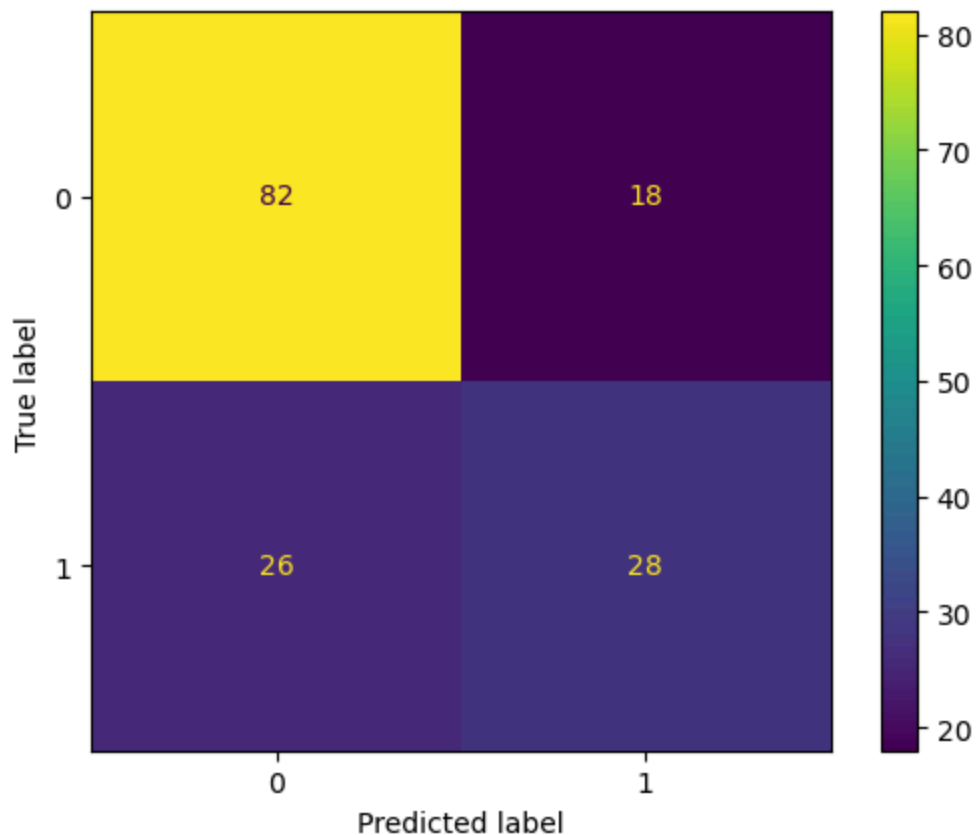
```
In [17]: print(classification_report(y_test, y_pred))
```

	precision	recall	f1-score	support
0	0.76	0.82	0.79	100
1	0.61	0.52	0.56	54
accuracy			0.71	154
macro avg	0.68	0.67	0.67	154
weighted avg	0.71	0.71	0.71	154

10. Confusion Matrix Visualization

The confusion matrix provides a visual representation of prediction outcomes, including false positives and false negatives. In a healthcare context, these errors are especially significant, as incorrect predictions may lead to delayed treatment or unnecessary medical intervention. This visualization supports the identification of potential bias patterns within model predictions.

```
In [18]: cm = confusion_matrix(y_test, y_pred)
disp = ConfusionMatrixDisplay(confusion_matrix=cm)
disp.plot()
plt.show()
```



11. Error Handling, Debugging, and Fairness Considerations

During initial testing, convergence warnings were encountered and resolved by increasing the maximum iteration limit. Feature scaling was applied to ensure that variables with larger magnitudes did not dominate model learning, which can contribute to biased outcomes.

Stratified sampling was used during train-test splitting to preserve class balance and reduce the risk of skewed predictions. These steps support both technical stability and fairness by promoting more consistent model behavior across patient groups.

12. Data Security, Privacy, and Ethical Compliance

The dataset used in this project contains no personally identifiable information (PII) and is used strictly for academic purposes. Data access was limited to the local development environment to prevent unauthorized modification or distribution.

Ethical considerations were integrated into the modeling process by prioritizing transparency, minimizing bias through preprocessing decisions, and evaluating error distributions that could impact healthcare equity. These practices align with responsible AI principles in healthcare.

13. Conclusion

This notebook demonstrates the implementation of a predictive healthcare model with an emphasis on reducing bias through careful model selection, preprocessing, and evaluation. By combining standard performance metrics with interpretability and fairness-aware analysis, the project highlights the importance of ethical machine learning practices in healthcare applications.

The results indicate that the model is functionally sound while providing a foundation for future fairness-enhancing techniques, such as subgroup performance analysis or bias mitigation algorithms.